Pattern Recognition 162 (2025) 111375



Contents lists available at ScienceDirect

Pattern Recognition

journal homepage: www.elsevier.com/locate/pr





Breaking boundaries: Low-precision conditional mutual information for efficient feature selection

Laura Morán-Fernández [a,*], Eva Blanco-Mallo [a], Konstantinos Sechidis [b], Verónica Bolón-Canedo [a]

[a] CITIC, Universidade da Coruña. A Coruña, Spain
[b] Advanced Methodology and Data Science, Novartis Pharma AG, Basel, Switzerland

ARTICLE INFO

Keywords:
Conditional mutual information
Low-precision
Feature selection
Markov blanket
Conditional mutual information maximization
IoT
Edge computing

ABSTRACT

As internet-of-things (IoT) devices proliferate, the need for efficient data processing at the network edge becomes increasingly critical due to the vast amounts of data generated. This paper presents a groundbreaking approach that leverages edge computing to address these challenges, using low-precision conditional mutual information (CMI) for feature selection. Our novel methodology improves the efficiency of edge computing systems by significantly reducing memory and energy consumption while maintaining high accuracy. We adapt this approach to feature selection algorithms, specifically, conditional mutual information maximization (CMIM) and incremental association Markov blanket (IAMB), and demonstrate its effectiveness for diverse datasets, including complex DNA microarrays. Our results show that low-precision methods not only compare competitively with traditional 64-bit implementations, but also yield significant performance and resource savings. For IoT and other machine learning applications, this work represents a significant advance in the development of more sustainable and efficient algorithms that can optimize computational resources and reduce their environmental impact.

1. Introduction

The proliferation of internet-of-things (IoT) devices and mobile computing has led to the generation of vast amounts of data that require efficient processing. This situation has highlighted the need to push the frontiers of machine learning to the network edge, so as to allow peripheral devices to analyze data generated on the device rather than transferring the data to the cloud for processing. This strategy, known as edge computing [1], not only optimizes resource usage by reducing network traffic but also enables real-time data analysis, which is crucial for applications requiring low latency and high-speed responses. However, traditional algorithms struggle to comply with the stringent memory and energy constraints of embedded systems, highlighting the need for more resource-efficient methods for machine learning tasks.

Despite significant advances in the processing capabilities of IoT devices, the memory and computational power limitations of the embedded systems still present substantial challenges. In this context, feature selection (FS) [2,3] needs to be adapted to operate in low-precision environments. FS plays a pivotal role in data analysis by reducing dimensionality, improving model interpretability, and enhancing computational efficiency. In high-dimensional datasets, selecting the most informative features is crucial to minimizing overfitting and ensuring robust model performance, especially in resource-constrained environments such as edge computing and IoT systems. In this paper we propose a novel methodology for FS in edge computing systems, based on low-precision conditional mutual information (CMI) [4].

The use of low-precision CMI not only significantly reduces storage requirements and energy consumption but also maintains a high level of FS accuracy. This accuracy is achieved by implementing lookup tables to compute the probability of events in low-precision representations. Furthermore, it has been demonstrated that a 16-bit representation can match the performance of a 64-bit representation in terms of feature ranking across various datasets [5].

Our work extends the application to FS of low-precision methods such as conditional mutual information maximization (CMIM) and incremental association Markov blanket (IAMB) [4]. Our extensive experimental study shows that these approaches are not only feasible but also comparable in terms of efficiency and effectiveness with traditional high-precision methods. The promising results of our low-precision

* Corresponding author.
E-mail addresses: laura.moranf@udc.es (L. Morán-Fernández), eva.blanco@udc.es (E. Blanco-Mallo), kostas.sechidis@novartis.com (K. Sechidis), vbolon@udc.es (V. Bolón-Canedo).

https://doi.org/10.1016/j.patcog.2025.111375
Received 1 July 2024; Received in revised form 26 September 2024; Accepted 12 January 2025
Available online 20 January 2025
0031-3203/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

CMI methods suggest that, in order to further optimize the trade-offs between accuracy, memory usage, and energy efficiency in edge computing environments, future work should explore implementation on specialized hardware, such as field-programmable gate arrays (FPGAs).

The main contributions of this paper are summarized as follows:

- We propose a new approach to computing CMI in low-precision environments, specifically adapted to the CMIM and IAMB algorithms for FS, thereby enabling FS in edge computing systems while significantly reducing memory and energy consumption without sacrificing accuracy.
- We demonstrate, thorough experiments performed with various datasets (including DNA microarrays), that low-precision versions (16- and 32-bit) achieve similar performance to standard 64-bit implementations in high-dimensional FS tasks.
- Leveraging low-precision methods, we demonstrate significant resource savings for IoT and machine learning applications, thereby ultimately rendering edge computing more efficient and sustainable in a way that reduces the carbon footprint of real-world systems.

The remainder of this paper is organized as follows: Section 2 reviews the state of the art in FS and CMI in low-precision environments, and Section 3 outlines specific methods used for CMI-based FS. Section 4 describes the proposed methodology for computing low-precision CMI, and Section 5 describes the experimental setup and results. Finally, Section 6 concludes with a discussion of the findings and suggestions for future research.

2. State of the art

Despite the growing need for efficient computing on edge devices, the literature offers limited insights into FS algorithms tailored for such contexts. One of the first studies to develop a specific procedure for low-precision environments was conducted by Morán-Fernández et al. [6], who proposed a mutual information (MI)-based algorithm with a limited number of bits for embedded systems that was tested on three different filtering methods relying on the MI criterion. Resource use was optimized using a lookup table, which stored the required values in terms of reduced precision. Their experimental results with various datasets showed that a 16-bit representation achieved the same feature ranking as a 64-bit representation. Additionally, the effectiveness of the procedure was demonstrated using fixed-point representations for limited-bit depth conditional mutual information (CMI) within a Markov blanket procedure [5]. Other FS approaches have focused on developing methods aimed at improving algorithm performance using data generated by small devices; however, they have not taken computational constraints into account in implementations. Nimbalkar and Kshirsagar [7] have proposed an FS method using information gain (IG) and gain ratio (GR) for intrusion detection in IoT systems, while Sun et al. [8] have described an FS approach based on the maximal information coefficient and a heuristic search strategy supported by an approximate Markov blanket algorithm. Via a Monte Carlo sampling approach, Chakraborty et al. [9] have developed an algorithm to handle FS under budget constraints, whereby a cross-entropy (CE)-based stochastic optimization technique converts the deterministic optimization problem into its stochastic counterpart, with traffic from real devices used to demonstrate the capability of their proposal with minimal misclassification risk.

Most other research focused on developing low-precision procedures tackles neural networks [10], both in terms of structure and optimization [11]. Particularly noteworthy are quantization, which uses reduced data types, and pruning, which involves removing any parameter with no impact on network accuracy [12]; both have been validated in terms of efficacy by numerous studies. In evaluating the performance of low-precision quantized neural networks (focusing specifically on LeNet and ConvNet architectures), Tatar et al. [13] reported that quantization significantly reduces the computational requirements and memory usage of neural networks, making them more suitable for deployment on resource-constrained devices. Roy et al. [14] have described a dynamic pruning-while-training method (integrating pruning directly into deep neural network training) that reduces computational and memory requirements with a minimal impact on accuracy. Zhou and Hu [15] discuss using evolutionary strategies to improve network initialization through re-initializing and re-pruning, showing how this method can enhance performance without increasing network size. Finally, Tung and Mori [16] have made an important contribution with their innovative approach to deep neural network compression, using a method that simultaneously applies pruning and quantization during training; their results demonstrate that this integrated approach not only preserves deep neural network accuracy but also enhances operational efficiency, making this a promising strategy for the deployment of complex models on small devices.

3. CMI for FS

CMI is a common concept in information theory that measures dependence between two variables when the state of a third variable is known. This criterion is defined in terms of expected Kullback–Leibler divergence as follows:

$$\begin{aligned} I(X;Y|Z) &= H(X|Z) - H(X|YZ) \\ &= \sum_{z\in Z} p(z) \sum_{x\in X} \sum_{y\in Y} p(xy|z) ln \frac{p(xy|z)}{p(x|z)p(y|z)} \end{aligned} \tag{1}$$

This can be conceptualized as the information shared between variables X and Y after the value of a third variable, Z, has been disclosed. Unlike conditional entropy, CMI can exceed unconditional MI, because of positive interaction between variables, where the correlation between two variables (X and Y) is enhanced on knowing of the state of a third variable (Z). CMI is zero when the two variables are independent, given the presence of the third variable. The maximum CMI value is the minimum of the two conditional entropies $H(X|Z)$ and $H(Y|Z)$, again obtained when the knowledge of one variable (and the conditioning variable) allows perfect prediction of the state of the other variable.

To compute CMI, we need the estimated distributions $\hat{p}(xy|z)$, $\hat{p}(x|z)$, and $\hat{p}(y|z)$. The probability of any particular event $p(x) = p(X = x)$ is estimated by maximum likelihood and the frequency of occurrence of an event $X = x$ is divided by the total number of events.

3.1. CMI-based FS methods

The CMI definition is useful in the FS context because it scores features following this criterion. Of the several FS methods based on CMI described in the literature, we use CMIM [17], as it captures relevancy, redundancy, and complementarity (each one desirable), and also provides a very good trade-off in terms of accuracy and stability [4].

Apart from CMIM, many other criteria can be derived from optimizing CMI, among them the Markov blanket discovery algorithms. The Markov blanket of a target Y is a set of features X_{MB} with the property $Y \perp\!\!\!\perp Z|X_{MB}$ for every $Z \subseteq X \setminus X_{MB}$. A set is referred to as a Markov boundary if it is a minimal Markov blanket, i.e., none of its subsets is a Markov blanket. In addition to playing a huge role in structure learning by Bayesian networks [18], the Markov blanket is also related to another key machine learning activity, namely, FS. Of the many different approaches to deriving the Markov blanket from finite datasets, we use IAMB [19].

3.1.1. CMIM

CMIM [17] is a popular multivariate FS method that, employing CMI to calculate relevancy and redundancy, selects features that maximize MI with their feature class to predict (Y), conditional on the response of any feature already selected (S):

$$CMIM(X_k) = min_{j \in S} I(X_k; Y|X_j) \tag{2}$$

This criterion selects features different from those already picked, ensuring a good trade-off between relevancy and redundancy: even if features already picked are individually significant, they do not carry more information about the class prediction. A higher CMIM value means feature X_k is relevant to target Y and is highly complementary with another selected feature X_j where $j \in S$.

3.1.2. IAMB

IAMB [19], a refinement of the grow-shrink algorithm [20], adopts the 2-stage structure common to many local learning algorithms: firstly, growing the candidate Markov blanket by adding new features until all remaining unselected features are conditionally independent of the target, then shrinking the candidate Markov blanket to remove any false positives that may have been selected (see Algorithm 1). Of the existing measures of association to decide which feature to be added to the candidate Markov blanket during the growing phase, the *significance of the conditional independence test* stands out as more appropriate in statistical terms than the raw CMI value [21]. To choose the most strongly related features, p-values for the conditional independence test play a crucial role in selecting features with smaller values.

Algorithm 1: IAMB

Input: Target Y, features $X = X_1, ..., X_d$, significance level α
Output: Markov blanket: X_{MB}

```
Phase I: forward - growing
X_CMB = ∅
while X_CMB has changed do
    Find X ∈ X \ X_CMB most strongly related with Y given X_CMB
    if X ⫫̸ Y|X_CMB using significance level α then
        Add X to X_CMB
Phase II: backward - shrinkage
foreach X ∈ X_CMB do
    if X ⫫ Y|X_CMB \ X using significance level α then
        Remove X from X_CMB
```

For IAMB, we need to test the conditional independence of X and Y given a subset of features Z. For fully observed categorical data we can use the G-test, a generalized likelihood ratio test, whereby the test statistic is calculated from sample data counts arranged in a contingency table. For more details, see Sechidis and Brown [22].

4. CMI adapted to low-precision environments

The methodology for adapting CMI to low-precision environments involves transforming CMI computation to efficiently use limited-bit representations, critical for edge computing systems with limited resources.

As mentioned in Section 3, computing CMI requires estimating probability distributions, which in turn involves counting the frequency of value occurrences within a given group. Building on work on approximating probabilities by Tschiatschek and Pernkopf [23], previous studies have explored MI and CMI using low-precision counters with a limited number of bits [5,24], and using fixed-point representations instead of the standard 64-bit floating-point resolution as used by hardware platforms. The motivation for this shift to fixed-point arithmetic is that the logic required for arithmetic operations performed in a 64-bit double-precision floating-point format consumes more silicon area and power than the same operation in a 32-bit fixed-point format. A 32-bit fixed-point adder, for instance, consumes about 9 times less power than its floating-point counterpart, according to research by Horowitz [25], while according to Dally [26], a fixed-point adder occupies up to 30 times less silicon area than a floating-point version.

Since CMI parameters are typically expressed in logarithmic terms, we compute the number of occurrences of an event and use a lookup table to determine the logarithm of the probability of a particular event. The lookup table, indexed in terms of the number of occurrences of individual events (individual counter) and the total number of events (total counter), stores values for the logarithms in the desired low-precision representation.

To limit the size of the lookup table and the bit width required for the counters, we assumed some maximum integer number M, where bi is the number of integer bits and where bf is the number of fractional bits. The lookup table L was pre-computed so that:

$$L(i,j) = \left[\frac{ln(i/j)}{q} \right]_R \cdot q, \tag{3}$$

where $[\cdot]_R$ denotes rounding to the nearest integer, q is the quantization interval of the desired fixed-point representation (2^{-bf}), and $ln(\cdot)$ denotes the natural logarithm, and where the counters i and j are in the range $\{0, \dots, M-1\}$. For more details on how this lookup table was calculated, see Morán-Fernández et al. [24].

Finally, to implement the low-precision method for evaluating the CMI measure, we applied the logarithmic product and quotient rules to Eq. (1), yielding the following expression:

$$\begin{aligned} I(X;Y|Z) = & \sum_{z \in Z} p(z) \sum_{x \in X} \sum_{y \in Y} p(xy|z)(ln(p(xy|z)) \\ & - (ln(p(x|z)) + ln(p(y|z)))) \end{aligned} \tag{4}$$

4.1. Empirical study

Before integrating our proposal into an FS procedure, we evaluated the effectiveness of the low-precision CMI measure, using 4, 8, 16, and 32 bits compared to the baseline 64-bit representation using synthetic data. Regarding independence, we used the G-test for a significance level of $\alpha = 0.05$ to test the conditional independence of X and Y given another variable Z, i.e., $X \perp\!\!\!\perp Y|Z$. To generate the data under this conditional independence, we used the following graph structure $X < -Z- > Y$, also known as a fork configuration [27]. We first randomly sampled Z, and then generated X and Y. Since the null hypothesis ($X \perp\!\!\!\perp Y|Z$) is true, we expect a false positive rate or type-I error rate of 5% [22]. For our experiments, we used different sample sizes (from n = 1000 to n = 10,000) and we assumed $|X| = 2$, $|Y| = 2$, and $|Z| = 10$.

Fig. 1 confirms our observations for the low-precision 32- and 16-bit methods, as well as the 64-bit baseline version, with a demonstrated false positive rate of 5% over 5000 repetitions. The small variations were due to the limited sample size. In particular, while the 8-bit approach failed to achieve the targeted type-I error rate, its consistency was evident, as 5% was approached with larger sample sizes. Conversely, the 4-bit approach showed very low false negative rates, as it reported underestimated values for CMI. The fact that this bit representation yielded negative values in 36% of the iterations is implausible, since the non-negativity property of CMI implies that $I(X;Y|Z)$ must always be greater than 0.

4.2. IAMB with low-precision CMI: p-value approximation

Under the null hypothesis that X and Y are statistically independent given Z, the G-statistic is known to be asymptotically χ^2-distributed, with $v = (|X| - 1)(|Y| - 1)|Z|$ degrees of freedom [28]. Thus, we can calculate the $\hat{p}_{XZ|Y}$ value as $1 - F(\hat{G})$, where F is the cumulative

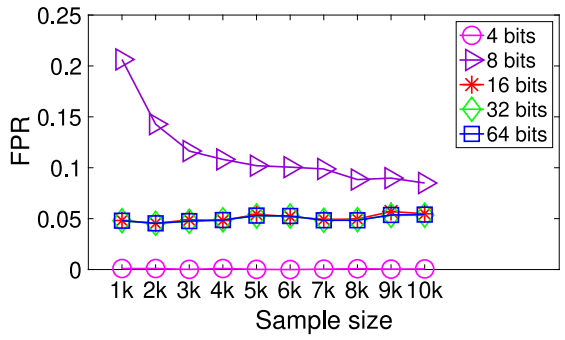


Fig. 1. Type-I error changes as a function of the number of samples, for a fixed $\alpha = 0.05$.

distribution function of the $\mathcal{X}^2$-distribution and $\hat{G}$ is the observed value of the G-statistic. The p-value represents the probability of obtaining a test statistic equal or more extreme than the observed one, given that the null hypothesis holds.

After calculating this value, we checked to see whether it exceeded a significance level α. If $p_{x,y,z} \leq \alpha$, we rejected the null hypothesis, and otherwise accepted it. This is the procedure followed to take the decision in lines 5 and 9 of Algorithm 1. In addition, to select the most strongly related feature in line 4, we evaluated the p-values and chose the features with smaller values.

The challenges associated with resource-constrained environments mean that performing the complex calculations required to obtain the p-value can lead to significant precision loss. To avoid this, we used an approximate table based on empirical data. The procedure to calculate the p-value involved selecting the most representative values required by the IAMB algorithm (Section 3.1.1) when processing a representative collection of data. Subsets of G-test values ($\hat{G}$) and degrees of freedom (v) were derived from empirical analyses of 60 real datasets, 70% for table construction and 30% for testing purposes. The datasets used, freely available online [29], vary widely in terms of both the number of instances and the number of features (in the ranges [21-67,557] and [8-12,533], respectively). This variability spans across binary to multi-class classification problems. Thus, our approximation was based on a broad spectrum of data, both in quantity and complexity terms.

Algorithm 2 condenses a collection of values into a representative subset. The algorithm takes a comprehensive list of observed values, $X = \{x_1, \ldots, x_x\}$, and a maximum desired size for the output set, m. This input list X represents either G-test statistics or degrees of freedom as observed across various datasets. The process starts by identifying all unique values within X, denoted as $U = \{u_1, \ldots, u_n\}$, and calculating their frequency, $C = \{c_1, \ldots, c_n\}$. If the number of unique values is less than or equal to the maximum size m, all those unique values are directly chosen for the output set, as they can all be considered representative and there is no need for further reduction. If the number of unique values is greater than m, the algorithm calculates an approximate grouping of size s by dividing the total count of values in C by m. This step aims to cluster the unique values into m groups based on their frequencies, thereby ensuring broad coverage with a limited number of representatives. In each iteration, the algorithm identifies a subset of values (U'), ensuring that their cumulative frequencies (C') are less than or equal to the approximate grouping size s. From this subset, the value with the highest frequency is selected as part of the output set. This iterative process continues until the cumulative frequency of the remaining elements in U is below the grouping size s or until there are no elements left.

After the required values were selected, the table was computed and stored at the required precision according to the following structure. A matrix T is linked to 2 vectors, G and V, that index the $\hat{G}$ and v values, respectively. The matrix itself contains pre-computed p-values

Algorithm 2: Pseudo-code to obtain the most relevant values

Input: List of values X and a maximum result size m
Output: Set of target values $edges$

```
1  U, C = unique(X) ;
2  if |U| < m then
3  |   edges ← U;
4  else
5  |   s = ∑(C)/m;
6  |   while U ≠ [] do
7  |   |   if ∑(C) ≥ s then
8  |   |   |   U', C' = U_i C_i | ∑ C_i ≤ s;
9  |   |   |   if U' ≠ [] then
10 |   |   |   |   edges ← u'_j | c'_j = max(C');
11 |   |   |   |   U, C = U_i C_i | ∑ C_i > s;
12 |   |   |   else
13 |   |   |   |   edges ← u'_j | j = 1;
14 |   |   |   |   U, C = U_i C_i | ∑ i > 1;
15 |   |   else
16 |   |   |   U = [];
17 return edges
```

for combinations of $\hat{G}$ and v, which are accessed as follows:

$$p - value(\hat{G}, v) = 1 - T(i, j), \tag{5}$$

where i is the index corresponding to the closest value in G to $\hat{G}$, and j is the index corresponding to the closest value in V to v. Fig. 2 depicts an example of how to obtain the vectors G and V to construct an approximate table from observed values when processing multiple datasets with the IAMB algorithm. In this process, Algorithm 1 is in charge of recording the values assumed by a given variable x during the execution of the algorithm through the different datasets. Subsequently, Algorithm 2 is responsible for extracting the most representative values from the recorded data. Finally, Fig. 3 shows how to obtain the p-value using a lookup table of size 3×3.

5. Experimental results

Below we describe an evaluation of our low-precision CMI approach in the context of an FS process, focusing on two popular CMI-based methods, namely, CMIM and IAMB. We performed two kinds of experiments (Fig. 4), using several datasets previously discretized by the equal-width strategy, which divided the range of values into 5 bins. The experiments for each FS method are as follows:

- **Feature selection quality**. In this first analysis, we studied the efficiency of the proposed low-precision FS methods. To do this, we analyzed the similarity of the features selected by the low-precision versions (4, 8, 16, and 32 bits), with the 64-bit version as our baseline. Note that, in terms of evaluating FS performance, CMIM and IAMB differ in that CMIM returns a ranking of features, whereas IAMB returns a feature subset. For both we calculated the true positive rate as a measure of the proportion of features that were correctly identified as such (regardless of their order of appearance) and using the 64-bit CMI version as the ideal output. This metric ranged from 0 to 1, where 1 indicated that the two outputs were identical, and 0 indicated that the outputs were completely dissimilar.
- **Classification performance**. After applying the CMIM and IAMB methods for FS, we investigated the potential impact of low-precision FS on classification, using a simple nearest neighbor (kNN) algorithm [30] with $k = 3$, as this makes minimal assumptions about the data and eliminates the need for parameter tuning.

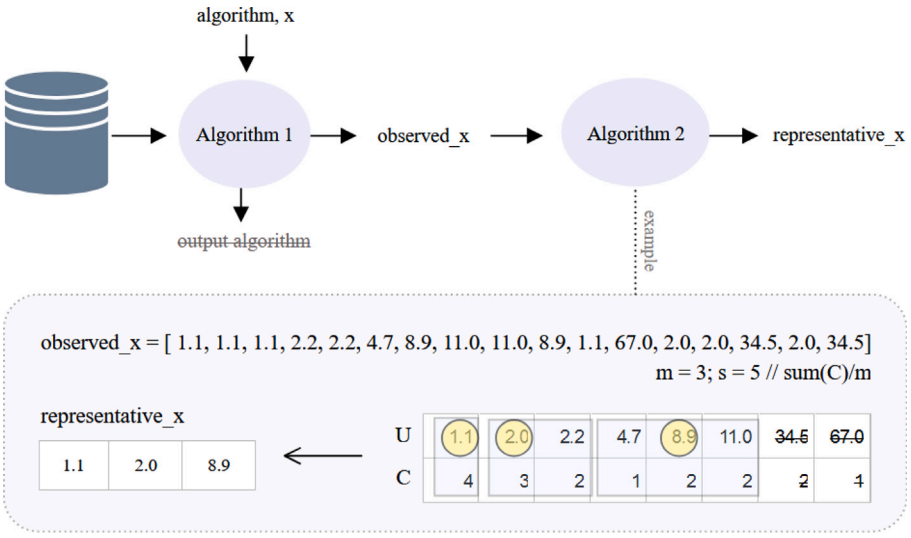


Fig. 2. Illustration of the procedure used to obtain the most representative values from observed data. This method was applied to derive the values of G and V for constructing the approximate table, based on the IAMB algorithm's processing of several datasets.

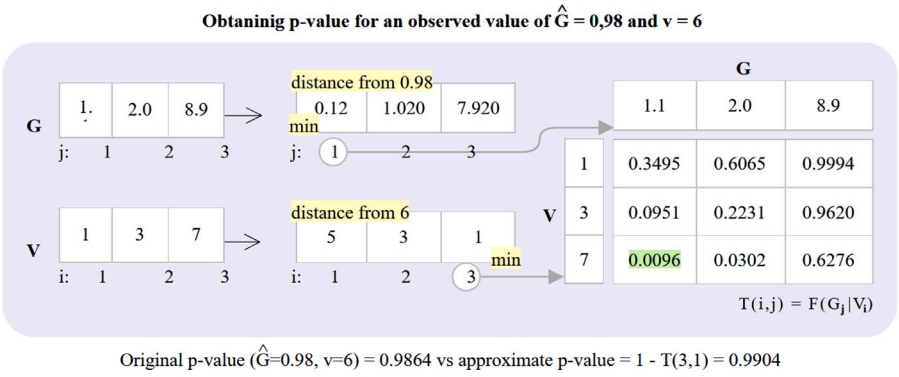


Fig. 3. Example of how to obtain a p-value for given values of $\hat{G}$ and v using the lookup table.

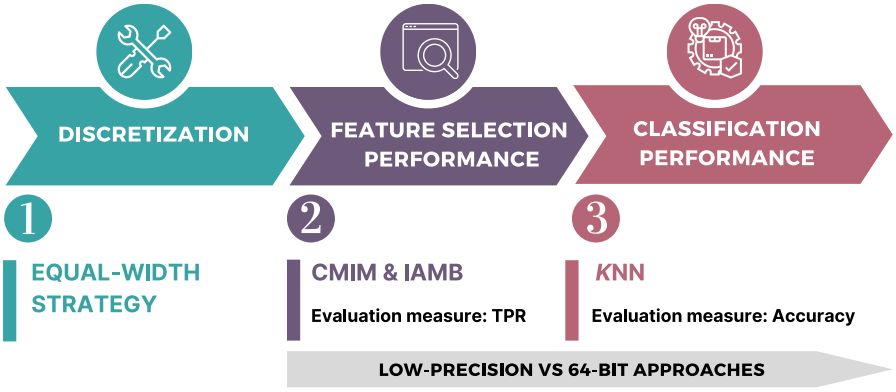


Fig. 4. Overview of the experiments (TPR, true positive rate).

Table 1
Main characteristics of the six datasets.

Dataset	#features	#samples	#classes
bc-wisc-diag	30	569	2
colon	2000	62	2
HARS	562	10,299	6
krvskp	36	3196	2
leukemia-1	5327	72	3
madelon	500	2400	2

At this point, it must be clarified that the inclusion of classifiers in our experiments may obscure the observations related to FS performance with limited bits, as these are associated with their own assumptions and nuances. It has been shown elsewhere that certain classifiers can achieve high accuracy levels even with suboptimal feature rankings [31]. To evaluate the error rate, we used a 3x5-fold cross-validation approach, which included both the FS and classification steps in a single loop.

5.1. Experimental setup

For the experiments, we considered six synthetic and real datasets (see Table 1), chosen to cover a wide range of sample-feature ratios and multi-class problems. Of the four datasets from the UCI and Kaggle repositories [29,32], *HARS* is of particular note, as it was curated using IoT devices. Additionally, to address the inherent complexity posed by high-dimension datasets, we included two DNA microarray datasets, *colon* and *leukemia-1* [33,34]. The experiments were performed in a Matlab2023b and Weka 3.8.6 environment [35].

5.2. CMIM application

Below we present and discuss our experimental results for CMIM in terms of feature selection performance and classification performance, comparing the different low-precision approximations and the baseline double precision (64-bit) approach.

5.2.1. Feature selection performance (ranking)

The first step to verify the efficiency of the proposed low-precision CMI was to analyze the similarity between the outputs obtained by the low-precision (4, 8, 16, and 32 bits) and 64-bit CMI after implementing the CMIM method. Since CMIM returns a ranking (not a subset like IAMB), as mentioned above, it requires a threshold, and for high dimensional applications, it is common to focus only on n-top features, e.g., $n = 5, 10$ and 20. Table 2 shows the true positive rates for the six datasets. As can be seen, the rankings obtained for the 16-bit and 32-bit approaches were practically the same as for the baseline 64-bit approach (i.e., values close to 1). The true positive rates were above 0.6 for the 8-bit approximation for all the datasets, but were significantly reduced for the 4-bit approach, where satisfactory performance was observed only for the smaller datasets (*krvskp* and *bc-wisc-diag*).

5.2.2. Classification performance

Table 3 shows classification accuracy and corresponding standard deviation of the low-precision methods compared to the 64-bit baseline method using CMIM in conjunction with *k*NN for three different sets of n-top features. Initial observations show a striking similarity between the results for the 8-, 16-, and 32-bit approximations and those for the 64-bit baseline. However, the 4-bit variant shows significantly lower precision values, especially for $n = 5$ features. Of particular note are the elevated standard deviation values observed for the *colon* and*leukemia-1* microarrays, rationalized by the extensive feature sets inherent to those datasets, which suggest a propensity for greater variance among selected features across iterations of the process.

Table 2
True positive rates for low-precision approaches (4, 8, 16 and 32 bits) using CMIM for three different n-top selected features.

Dataset	n-top features	4-bit	8-bit	16-bit	32-bit
bc-wisc-diag	5	0.00	0.80	1.00	1.00
	10	0.20	0.80	1.00	1.00
	20	0.75	0.85	0.95	1.00
colon	5	0.00	0.80	1.00	1.00
	10	0.10	0.80	1.00	1.00
	20	0.15	0.75	1.00	1.00
HARS	5	0.00	0.60	1.00	1.00
	10	0.10	0.40	1.00	1.00
	20	0.30	0.60	0.95	1.00
krvskp	5	0.80	0.80	1.00	1.00
	10	0.70	0.60	0.90	1.00
	20	0.60	0.70	0.95	1.00
leukemia-1	5	0.20	0.60	0.80	1.00
	10	0.10	1.00	1.00	1.00
	20	0.20	0.90	1.00	1.00
madelon	5	0.20	0.80	1.00	1.00
	10	0.10	0.60	1.00	1.00
	20	0.25	0.45	0.70	1.00

Although not the primary focus of this study, it is pertinent to acknowledge that, with the exception of the *madelon* dataset, maximum accuracy is achieved for larger feature selections, most notably $n = 20$ features. Note that the synthetic *madelon* dataset, originally proposed in the 2003 NIPS Feature Selection Challenge, has its five relevant features at the vertex of a 5-dimensional hypercube.

To assess the statistical significance of our classification results, we used a Friedman test followed by a Nemenyi post-hoc test to examine the classification error. Fig. 5 shows critical difference (CD) diagrams, as proposed by Demšar (2006) to illustrate the relationship between groups of methods that do not show significant differences (at α=0.05). The primary axis of the CD diagrams denotes the average ranking of the methods, with axis orientation adjusted so that lower (better) ranks appear to the right.

The analysis shows that for smaller subsets of features ($n = 5$ and $n = 10$), the 64-, 32-, 16-, and 8-bit versions consistently outperform the 4-bit model to a statistically significant degree. However, when considering a larger set of presumably relevant features ($n = 20$), the 4-bit version is the least effective, although this difference lacks statistical significance.

5.3. IAMB application

Below we evaluate low-precision CMI using IAMB. As mentioned, the output of this method is a subset of the original features, not a ranking.

5.3.1. Feature selection performance (subset)

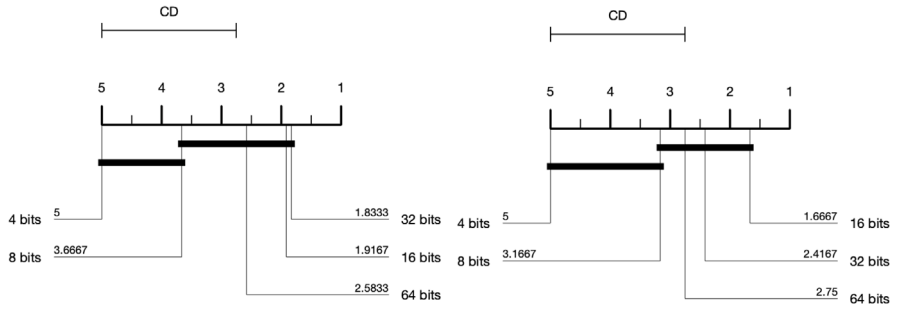
The true positive rate measured accuracy in identifying features, with the baseline IAMB (64 bits) serving as the optimal feature subset. As shown in Table 5, low-precision approaches using 16 and 32 bits tended to select the same features as the 64-bit version. In contrast, as expected from the results of the empirical study (Section 4.1), the 4-bit approach failed to capture any features identified by the 64-bit version. In particular, for the *HARS* dataset, since the low-precision approach yielded an empty feature set, we excluded this low-precision size from subsequent experiments.

5.3.2. Classification performance

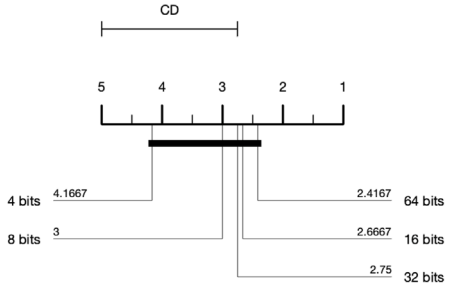
Table 4 compares the 8-bit, 16-bit, 32-bit, and 64-bit versions for classification, showing accuracy results and standard deviations for the 6 datasets after applying IAMB and the *k*NN classifier.

Table 3
Classification accuracy and standard deviations for the low-precision approaches (4, 8, 16 and 32 bits) and the baseline approach (64 bits) using CMIM and kNN for three different n-top selected features (5, 10, and 20).

Dataset	n-top features	4-bit	8-bit	16-bit	32-bit	64-bit
bc-wisc-diag	5	93.50 ± 1.12	95.60 ± 0.81	95.84 ± 0.54	95.72 ± 1.11	95.72 ± 1.11
	10	95.31 ± 1.67	95.43 ± 0.70	95.84 ± 0.66	95.90 ± 0.66	95.90 ± 0.66
	20	96.13 ± 1.05	95.72 ± 0.95	96.19 ± 0.32	96.54 ± 0.49	96.54 ± 0.49
colon	5	69.96 ± 7.41	75.94 ± 4.56	77.01 ± 5.58	77.56 ± 5.29	75.26 ± 3.38
	10	73.72 ± 5.50	80.81 ± 6.22	79.14 ± 4.93	78.12 ± 4.04	77.01 ± 3.86
	20	79.53 ± 4.21	84.10 ± 4.33	83.31 ± 3.89	82.31 ± 3.89	81.20 ± 4.19
HARS	5	83.49 ± 2.93	90.64 ± 0.46	90.74 ± 0.17	90.74 ± 0.17	90.74 ± 0.17
	10	91.96 ± 0.50	92.46 ± 0.44	92.62 ± 0.35	92.53 ± 0.28	92.53 ± 0.28
	20	93.41 ± 0.25	93.10 ± 0.34	93.11 ± 0.16	93.25 ± 0.32	93.25 ± 0.32
krvskp	5	79.72 ± 4.31	91.90 ± 0.49	93.33 ± 0.84	93.33 ± 0.84	94.09 ± 0.47
	10	87.39 ± 4.66	93.52 ± 0.56	94.66 ± 0.39	94.53 ± 0.39	94.57 ± 0.29
	20	94.36 ± 0.75	94.73 ± 0.93	94.68 ± 0.52	94.68 ± 0.23	94.77 ± 0.28
leukemia-1	5	81.94 ± 1.26	90.89 ± 2.64	92.73 ± 2.84	93.21 ± 2.84	89.87 ± 2.36
	10	85.30 ± 4.71	90.51 ± 5.73	90.51 ± 5.73	90.51 ± 5.73	89.94 ± 3.95
	20	83.97 ± 4.14	92.19 ± 3.06	92.73 ± 2.98	92.25 ± 3.38	92.97 ± 1.79
madelon	5	69.24 ± 3.91	84.22 ± 2.35	88.18 ± 0.65	88.21 ± 0.77	88.21 ± 0.77
	10	70.42 ± 1.62	91.30 ± 0.94	92.97 ± 0.16	92.94 ± 0.30	92.94 ± 0.30
	20	69.18 ± 3.12	86.46 ± 1.34	86.08 ± 1.41	86.06 ± 0.60	86.06 ± 0.60



(a) 5-top features (b) 10-top features



(c) 20-top features

Fig. 5. Critical difference diagrams showing average ranking after applying CMIM to the four low-precision approaches (4, 8, 16 and 32 bits) and the baseline approach (64 bits).

Table 4
True positive rates for the low-precision approaches using IAMB, with an empty feature subset returned for *HARS*.

Dataset	4-bit	8-bit	16-bit	32-bit
bc-wisc-diag	0.00	1.00	1.00	1.00
colon	0.00	1.00	1.00	1.00
HARS	–	0.25	1.00	0.75
krvskp	0.37	0.62	0.87	0.87
leukemia-1	0.00	1.00	1.00	1.00
madelon	0.00	0.50	0.75	1.00

Table 5
Classification accuracy and standard deviations for the low-precision approaches (8, 16, and 32 bits) and the baseline approach (64 bits) for IAMB and kNN.

Dataset	8-bit	16-bit	32-bit	64-bit
bc-wisc-diag	88.39 ± 8.85	92.56 ± 0.84	92.56 ± 0.84	92.91 ± 0.82
colon	67.09 ± 4.60	70.98 ± 3.55	70.98 ± 3.55	70.98 ± 3.55
HARS	37.43 ± 4.04	62.49 ± 2.65	60.50 ± 2.38	66.54 ± 2.51
krvskp	94.10 ± 0.37	94.83 ± 0.38	95.26 ± 0.82	96.83 ± 0.99
leukemia-1	73.62 ± 7.58	70.44 ± 7.51	68.54 ± 6.85	75.49 ± 6.29
madelon	62.56 ± 4.83	72.14 ± 1.53	73.71 ± 2.89	84.12 ± 1.74

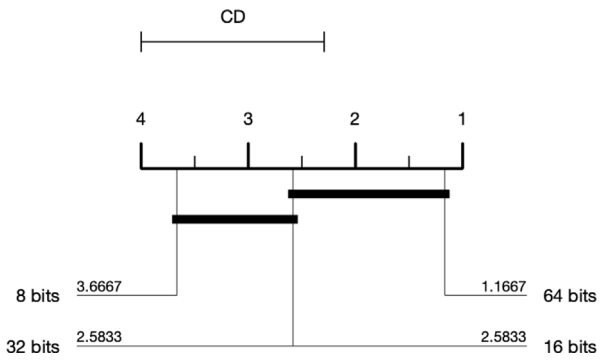


Fig. 6. Critical difference diagram showing the average outputs after applying IAMB to the low-precision approaches (8, 16, and 32 bits) and the baseline approach (64 bits).

The 8-bit approach performed poorly, especially when applied to the *HARS* dataset, which happens to have the largest number of samples. In terms of optimal performance, the baseline 64-bit version performed best, although note that its accuracy value was close to that of the 16-bit and 32-bit counterparts for most of the datasets.

In terms of standard deviation measurements, a pattern similar to that for the CMIM method is apparent. The low-precision version with fewest bits produced significantly higher standard deviation values than the alternative versions, and the standard deviation values tended to be more pronounced in datasets with a greater number of features, namely, the *colon* and *leukemia-1* microarrays.

To examine the statistical significance of our classification results, as done for CMIM, we performed a Friedman test in conjunction with a Nemenyi post-hoc test. The CD diagram in Fig. 6 shows that there were no significant differences between the 64-bit, 32-bit, and 16-bit versions.

Those results suggest that while outputs vary when there are fewer bits, this variation does not significantly affect classification accuracy—the primary metric for evaluating feature classification methods.

6. Conclusions

Our work contributes to the development of more sustainable and resource-efficient algorithms for real-world deployment. The described FS methodology evaluates the effectiveness of using CMI with low-precision parameters, integrated into the CMIM and IAMB algorithms, and incorporating a method for approximating the required p-value for the latter. The results of our evaluation for 6 different datasets highlight the successful performance of low-precision versions of both the CMIM and IAMB methods for FS compared to their 64-bit counterparts, even when dealing with complex datasets such as DNA microarrays. This suggests that the use of low-precision FS methods is a viable strategy for minimizing energy consumption and carbon emissions by IoT devices, and for improving computational efficiency and reducing resource overheads in various machine learning applications.

Despite the existence of tools such as the Green Algorithms carbon footprint calculator [36] and more specialized low-level tools such as Intel's Running Average Power Limit feature, the time and energy measurements for our experiment remain imprecise. Although the new implementation closely approximates fixed-point arithmetic, it introduces additional overheads to operations because the high-level simulation uses double-precision floating-point values for low-level operations.

Imperative for future research is exploration of the implementation of FS methods on FPGAs, as this would enable accurate measurement of time and memory usage (contrasting with our simulation-based approach where accuracy measurement is limited). Implementation on FPGAs would not only increase the realism of simulations, but would also help address the issue of high-level implementation overheads, making the research more applicable to real-world environments.

CRediT authorship contribution statement

Laura Morán-Fernández: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Eva Blanco-Mallo:** Writing – original draft, Visualization, Software, Methodology, Investigation. **Konstantinos Sechidis:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Verónica Bolón-Canedo:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by Xunta de Galicia/FEDER-UE under Grant ED431G 2023/01; Ministerio de Ciencia e Innovación MCIN/AEI/10.13039/501100011033 and 'Next-GenerationEU'/PRTR under Grants [PID2019-109238GB-C22; TED2021-130599A-I00; PID2023-147404OB-I00] and Ministry for Digital Transformation and Civil Service and 'Next-GenerationEU'/PRTR under grant TSI-100925-2023-1. CITIC, as a Research Center of the University System of Galicia, is funded by Consellería de Educación, Universidade e Formación Profesional of the Xunta de Galicia, Spain through the European Regional Development Fund (ERDF/FEDER) and the Secretaría Xeral de Universidades (ED431G 2023/01).

Data availability

The data used were downloaded from public repositories whose links are provided in the paper.

References

[1] B. Varghese, N. Wang, S. Barbhuiya, P. Kilpatrick, D.S. Nikolopoulos, Challenges and opportunities in edge computing, in: 2016 IEEE International Conference on Smart Cloud (SmartCloud), IEEE, 2016, pp. 20–26.
[2] I. Guyon, S. Gunn, M. Nikravesh, L.A. Zadeh, Feature Extraction: Foundations and Applications, vol. 207, Springer, 2008.
[3] S.H. Hammed, A. Albahri, Unlocking the potential of autism detection: integrating traditional feature selection and machine learning techniques, Appl. Data Sci. Anal. 2023 (2023) 42–58.
[4] G. Brown, A. Pocock, M.-J. Zhao, M. Luján, Conditional likelihood maximisation: a unifying framework for information theoretic feature selection, J. Mach. Learn. Res. 13 (1) (2012) 27–66.
[5] L. Morán-Fernández, E. Blanco-Mallo, K. Sechidis, A. Alonso-Betanzos, V. Bolón-Canedo, When size matters: Markov blanket with limited bit depth conditional mutual information, in: IoT Streams for Data-Driven Predictive Maintenance and IoT, Edge, and Mobile for Embedded Machine Learning, Springer, 2020, pp. 243–255.
[6] L. Morán-Fernández, V. Bolón-Canedo, A. Alonso-Betanzos, Low-precision feature selection on microarray data: an information theoretic approach, Med. Biol. Eng. Comput. 60 (5) (2022) 1333–1345.
[7] P. Nimbalkar, D. Kshirsagar, Feature selection for intrusion detection system in Internet-of-Things (IoT), ICT Express 7 (2) (2021) 177–181.
[8] G. Sun, J. Li, J. Dai, Z. Song, F. Lang, Feature selection for IoT based on maximal information coefficient, Future Gener. Comput. Syst. 89 (2018) 606–616.
[9] B. Chakraborty, D.M. Divakaran, I. Nevat, G.W. Peters, M. Gurusamy, Cost-aware feature selection for IoT device classification, IEEE Internet Things J. 8 (14) (2021) 11052–11064.
[10] T. Chu, Q. Luo, J. Yang, X. Huang, Mixed-precision quantized neural networks with progressively decreasing bitwidth, Pattern Recognit. 111 (2021) 107647.

[11] T. Liang, J. Glossner, L. Wang, S. Shi, X. Zhang, Pruning and quantization for deep neural network acceleration: A survey, Neurocomputing 461 (2021) 370–403.
[12] M. Neseem, C. McCullough, R. Hsin, C. Leichner, S. Li, I.S. Chong, A. Howard, L. Lew, S. Reda, V.-M. Rautio, et al., Pikelpn: Mitigating overlooked inefficiencies of low-precision neural networks, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 15996–16005.
[13] G. Tatar, S. Bayar, İ. Çiçek, Performance evaluation of low-precision quantized LeNet and ConvNet neural networks, in: International Conference on INnovations in Intelligent SysTems and Applications, IEEE, 2022, pp. 1–6.
[14] S. Roy, P. Panda, G. Srinivasan, A. Raghunathan, Pruning filters while training for efficiently optimizing deep learning networks, in: 2020 International Joint Conference on Neural Networks, IJCNN, IEEE, 2020, pp. 1–7.
[15] R. Zhou, T. Hu, Evolving better initializations for neural networks with pruning, in: Proceedings of the Companion Conference on Genetic and Evolutionary Computation, 2023, pp. 703–706.
[16] F. Tung, G. Mori, Deep neural network compression by in-parallel pruning-quantization, IEEE Trans. Pattern Anal. Mach. Intell. 42 (03) (2020) 568–579.
[17] F. Fleuret, Fast binary feature selection with conditional mutual information, J. Mach. Learn. Res. 5 (9) (2004).
[18] J.-P. Pellet, A. Elisseeff, Using Markov blankets for causal structure learning, J. Mach. Learn. Res. 9 (Jul) (2008) 1295–1342.
[19] I. Tsamardinos, C.F. Aliferis, Towards principled feature selection: relevancy, filters and wrappers., in: AISTATS, 2003.
[20] D. Margaritis, S. Thrun, Bayesian network induction via local neighborhoods, in: Advances in Neural Information Processing Systems, 2000, pp. 505–511.
[21] S. Yaramakala, D. Margaritis, Speculative Markov blanket discovery for optimal feature selection, in: Fifth IEEE International Conference on Data Mining, ICDM'05, IEEE, 2005, pp. 4–pp.
[22] K. Sechidis, G. Brown, Simple strategies for semi-supervised feature selection, Mach. Learn. 107 (2) (2018) 357–395.
[23] S. Tschiatschek, F. Pernkopf, Parameter learning of Bayesian network classifiers under computational constraints, in: Joint European Conference on Machine Learning and Knowledge Discovery in Databases, Springer, 2015, pp. 86–101.
[24] L. Morán-Fernández, K. Sechidis, V. Bolón-Canedo, A. Alonso-Betanzos, G. Brown, Feature selection with limited bit depth mutual information for portable embedded systems, Knowl.-Based Syst. 197 (2020) 105885.
[25] M. Horowitz, 1.1 computing's energy problem (and what we can do about it), in: 2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers, ISSCC, IEEE, 2014, pp. 10–14.
[26] W. Dally, High-performance hardware for machine learning, Nips Tutor. 2 (2015) 3, [Online]. https://berkeley-deep-learning.github.io/cs294-dl-f16/slides/DL_HW_Berkeley_0916.pdf.
[27] J. Pearl, M. Glymour, N.P. Jewell, Causal Inference in Statistics: A Primer, John Wiley & Sons, 2016.
[28] A. Agresti, M. Kateri, Categorical Data Analysis, Springer, 2011.
[29] K. Bache, M. Linchman, UCI Machine Learning Repository, University of California, Irvine, School of Information and Computer Sciences, 2024, URL http://archive.ics.uci.edu/ml/ (Accessed January, 2024).
[30] D.W. Aha, D. Kibler, M.K. Albert, Instance-based learning algorithms, Mach. Learn. 6 (1) (1991) 37–66.
[31] V. Bolón-Canedo, N. Sánchez-Maroño, A. Alonso-Betanzos, A review of feature selection methods on synthetic data, Knowl. Inf. Syst. 34 (3) (2013) 483–519.
[32] G.L. Kaggle, Kaggle datasets, 2024, URL https://www.kaggle.com/datasets. (Accessed January, 2024).
[33] Kent Ridge Biomedical Data Set Repository, 2024, URL https://leo.ugr.es/elvira/DBCRepository. (Accessed January, 2024).
[34] A. Statnikov, I. Tsamardinos, Y. Dosbayev, C.F. Aliferis, GEMS: a system for automated cancer diagnosis and biomarker discovery from microarray gene expression data, Int. J. Med. Informatics 74 (7–8) (2005) 491–503.
[35] K. Sechidis, Simple strategies for semi-supervised feature selection, 2018, https://github.com/sechidis [Accessed: January 2024].
[36] L. Lannelongue, J. Grealey, M. Inouye, Green algorithms: quantifying the carbon footprint of computation, Adv. Sci. 8 (12) (2021) 2100707.

Laura Morán-Fernández received her B.S. (2015) and Ph.D. (2020) degrees in Computer Science from the University of A Coruña (Spain). She is currently an Assistant Professor in the Department of Computer Science and Information Technologies of the University of A Coruña. She received the Frances Allen Award (2021) from the Spanish Association of Artificial Intelligence. Her research interests include machine learning, feature selection and green algorithms. She has co-authored five book chapters, and more than 20 research papers in international journals and conferences.

Eva Blanco-Mallo received her B.A. degree (2012) in Publicity and Public Relations from the University of Vigo (Spain) and her B.S. degree (2019) in Computer Sciences (2019) from the University of A Coruña (Spain). She is currently a Ph.D. student in the Department of Computer Science and Information Technologies of the University of A Coruña. Her research interest includes machine learning, transfer learning and recommender systems.

Konstantinos Sechidis received his B.S. degree in Electrical and Computer Engineering (2006) from the Aristotle University of Thessaloniki (Greece), M.S. degree in Communications and Signal Processing (2009) from the Imperial College London (UK), M.S. degree in Information Systems (2011) from the Aristotle University of Thessaloniki (Greece) and Ph.D. degree in Machine Learning (2015) from the University of Manchester (UK). He currently works in the Advanced Methodology and Data Science group at Novartis and he is an honorary research fellow in Machine Learning and Robotics within the Department of Computer Science in the University of Manchester. He is a member of the editorial board of Machine Learning Journal (MLJ) and vice-chair of the technical committee on Statistical Pattern Recognition Techniques of the International Association for Pattern Recognition (IAPR).

Verónica Bolón-Canedo received her B.S. (2009), M.S. (2010) and Ph.D. (2014) degrees in Computer Science from the Universidade da Coruña (Spain). After a postdoctoral fellowship in the University of Manchester, UK (2015), she is currently an Associate Professor in the Department of Computer Science and Information Technologies of the Universidade da Coruña. Her main current areas are machine learning and feature selection. She is co-author of more than 100 papers on these topics in international conferences and journals. She is member of the Young Academy of Spain and the Royal Academy of Sciences of Spain.